WORKSHEET

Chapter 9: Sample proportion probabilities

1 What percentage of values of $\hat{p}$ would lie between 0.85 and 0.95 for samples with $n = 40$ and $p = 0.8$?

2 What percentage of values of $\hat{p}$ would lie between 0.11 and 0.28 for samples with $n = 60$ and $p = 0.3$?

3 What percentage of values of $\hat{p}$ would lie between 0.58 and 0.82 for samples with $n = 120$ and $p = 0.7$?

4 What proportion of values of $\hat{p}$ would lie between 0.69 and 0.72 for samples with $n = 400$ and $p = 0.68$?

5 What proportion of values of $\hat{p}$ would lie between 0.28 and 0.42 for samples with $n = 75$ and $p = 0.32$?

6 The probability of an adult having blue eyes is 0.24. What is the probability that a sample of 300 adults has between 60 and 78 people with blue eyes?

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7 The ABS has found that 21% of Australians are Queenslanders and of these, 47% live in Brisbane.

a From 500 Australians, what is the probability that between 80 to 100 are Queenslanders?

b From 500 Queenslanders, what is the probability that between 210 and 250 live in Brisbane?

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Answers

1 20.6%

2 36.8%

3 99.6%

4 73.9%

5 29.1%

6 73.4%

7 **a** 28.8%

b 89.8%

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